

MATHEMATICS
Introduction to Modern Algebra I
Solutions to Quiz 6.

1) Let $\phi : G \rightarrow G'$ be a group homomorphism. Show that:

a) The image of ϕ is a subgroup of G' .

Pf. Closed under $*$: If g', h' are in the image of ϕ , then $g' = \phi(g)$ and $h' = \phi(h)$ for some $g, h \in G$. But then $g'h' = \phi(g)\phi(h) = \phi(gh)$ is in the image of ϕ .

Closed under inverses: If g' is in the image of ϕ , then $g' = \phi(g)$ for some $g \in G$. But then $g^{-1} \in G$ and $\phi(g)^{-1} = \phi(g^{-1})$ is in the image of ϕ .

b) The kernel of ϕ is a subgroup of G .

Pf. Closed under $*$: If g, h are in the kernel, then $\phi(g) = e'$ and $\phi(h) = e'$. But then $\phi(gh) = \phi(g)\phi(h) = e'e' = e'$, so that gh is also in the kernel.

Closed under inverses: If g is in the kernel, then $\phi(g) = e'$ and so $\phi(g^{-1}) = \phi(g)^{-1} = (e')^{-1} = e'$ so that g^{-1} is in the kernel.

c) The kernel of ϕ is a normal subgroup of G .

Let $K = \{g \in G \mid \phi(g) = e'\}$. We must show that $gKg^{-1} \subset K$ for all $g \in G$. Let $k \in K$, then $\phi(k) = e'$ and so $\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g^{-1}) = \phi(g)e'\phi(g)^{-1} = e'$ i.e. $gkg^{-1} \in K$.